



4D-BIM based dynamic construction site layout planning

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Received: 5 July 2025 / Accepted: 19 July 2025

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Abstract

Construction site layout planning is an activity concerned with placing temporary facilities within the construction site so as to achieve the highest possible efficiency of construction operations and thus reduce the cost of construction. This can be achieved by creating a dynamic site layout planning system that changes as construction requirements change. In this research, we introduce a 4D-BIM-based dynamic site layout planning model. To achieve the previous objective, the existing models that dealt with the site layout planning problem were reviewed. This helped identify the limitations of available models and suggestions for developing a practical site layout planning methodology. The main features of the site layout planning problem were discussed. An integrated model for solving the dynamic site layout problem using BIM technology was developed. The developed model was integrated with a schedule to develop a dynamic model. The validity of the proposed model was tested, and the strength and limitations of the proposed model were illustrated. The automated system demonstrated a dynamic approach that considers both the transition from one construction phase to another and the delaying and floating of schedule activities. Facilities are arranged continuously, where space is modeled in a coordinate system. The calculation of material quantities according to changes in the schedule and progress is facilitated by using BIM technology. Ultimately, this leads to a reduction in the effort and time spent on solving the problem, providing a more realistic solution that considers progress in construction and delays, and increasing the number of locations available for placing temporary facilities. This, in turn, improves the required optimized solution.

Keywords Building information modeling (BIM) · Dynamic construction site layout planning · Temporary facilities · Occupational safety in construction sites · Artificial intelligence · Genetic algorithms

1 Introduction

Construction site layout planning has gained a lot of attention as a new way to reduce cost, time as well as improve safety on construction sites, especially in large projects, where a large number of workers and equipment are involved in construction. The problem of layout planning is defined in general as the process of determining the number, type and size of temporary facilities required for construction as well as their locations within the construction site and their locations relative to each other so as to achieve the highest efficiency of construction without violating the imposed restrictions [1]. A detailed site layout planning absence might lead to various problems such as materials stacked in a wrong location, insufficient space, and unpractical facilities locations for construction operations [2]. Temporary facilities are those that serve on the construction site without being part of the structure to be constructed. The site layout problem background could be broken down

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into five sub-titles: (1) Time dimension in site layouts planning. (2) BIM technology usage in Site layout planning. (3) Site layout constraints. (4) Optimization methods. (5) Representation of the data.

This research methodology involves reviewing previous models for solving the construction site layout planning problem, discussing its main features, then determining a scientific methodology for construction site layout planning based on BIM technology. The developed model was then integrated with a schedule to develop a dynamic model. Finally, the validity of the proposed model is tested, and the strength of the proposed model and its limitations are clarified.

2 Literature review

Site layout models fall into two categories regards to time dimension (1) static layout models, which is interested in searching for the optimal location for each temporary facility, regardless of the duration of its need for the site. And (2) dynamic layout models, which, on the contrary, take into account the duration required for the facility to exist within the site [3]. Dynamic layout models outweigh the static approach in that site layout changes over time. Some dynamic models suggested dividing the project's duration into phases, and making a partial layout for each phase, whereby the planner's experience is relied upon to determine the length of each phase in a way that suits the nature of each project. In each partial layout, the locations of the temporary facilities are improved, so that some facilities remain fixed from one phase to the next, and the locations of the rest of the facilities are improved [4]. Other models suggested considering the actual period the facilities occupy the site space [5].

After years of gradual development, a major boom occurred in the construction industry, which contributed to this great boom Building Information Modeling (BIM), BIM is a digital representation of the physical and functional characteristics of a structure, serving as an informational resource about the structure throughout its lifecycle [6]– [7]. In BIM-based layout models, building data and necessary information to determine the number and size of temporary facilities are read from the BIM model automatically (such as the amount of different materials for columns, beams, or walls), which saves a lot of manual work especially in the case of dynamic layout planning. However, other information such as material frequency and schedule data require special design [8], where the schedule data is stored in a separate file and combined with the BIM model to produce a 4D-BIM model that can be used in the dynamic layout planning of the construction site. Some researchers

tried to take advantage of the dimensions of the structure components provided by the BIM model as a benchmark and compare them with the point cloud data in order to improve the tolerance control [9], developing a BIM-based prototype system which produces 3D layout and provides the necessary data for layout simulation [10], as well as suggesting a BIM-based method for estimating the frequency between facilities and buildings [8]. Some researchers have proposed intelligent systems for dynamic layout planning, for example integration of scheduling software and BIM-based design software to automate site layout planning [3], combining BIM and RFID (Radio Frequency Identification) system to detect available locations in real time to automate site layout planning [11], As well as proposing a dynamic model of materials requirements planning based on BIM to specify the required quantities and types of building materials [12]. In 2022 some researchers provided a model utilizes information from (BIM) and Geospatial Information System (GIS) for optimizing dynamic site layout [13]. Recently an improved multi-objective optimization model combining ant colony algorithm and Pareto solutions reduced computation time by 85.9%, significantly enhancing efficiency [14].

Site layout planning constraints are a set of logical rules and regulations that must be taken into account when planning site layouts [4]. One of the reasons for poor and impractical CSLP models is insufficient attention to site layout constraints [15]. In other words, the constraints must be properly defined and thoroughly checked for automated planning models to produce practical layouts [16]. However, current models of CSLP automation do not properly account for constraints as construction site planners do. Despite site boundary restrictions that do not allow any facility to fall outside the site boundaries, as well as overlapping restrictions that do not allow two facilities to overlap together are widely used, safety restrictions and environmental problems such as the possibility of falling objects and noise problems [17], and restrictions that concern leaving a safe distance between temporary facilities and neighboring buildings is rarely used [18]. Neglecting the CSLP constraints may lead to many problems such as inefficient travel path as well as increased costs due to relocation of some temporary facilities to achieve efficient construction, and in the end, it has a negative impact on optimizing construction site layout. Therefore, it is necessary to pay attention to the definition and application of various construction site planning constraints.

Optimization methods in site layout planning fall into two categories, exact and heuristic. Exact methods (such as sublevel algorithm, simple and finite level algorithm, cutting level algorithm, etc.) provide an accurate ideal solution for a single or multi-objective function, but they require a lot of effort and are best used for small projects. While heuristic

methods (such as genetic algorithms, particle swarm algorithms, ant colony algorithm, etc.) provide approximate solutions close to the global optimal one, but they fail to provide the global optimal solution. The heuristic methods, in contrast to the exact ones, are characterized by their fastness, lack of required effort, and their applicability to large-scale projects [19]. There are also hybrid methods, wherein both the heuristic and the exact methods are used in one combined model. Genetic algorithms are an artificial intelligence technique that performs a random search to find solutions close to optimal, based on the concept of natural selection and survival of the fittest [20–23]. Genetic algorithms have been widely used in the formation of site layout models, the first to use them in site layout problems are Li & Love [24]. Then Hegazy & Elbeltagi followed him, as they verified the model on a real case study [25]. Mawdesley, Yang, and Al-Jibouri used a sequence-based genetic algorithm to improve the model search technique. They further mapped the travel path by Euclidean route and graph theory [26]. In parallel Cheung, Tong, & Tam used a steady-state genetic algorithm model in which the parents were selected with a rank-based approach [27]. Moreover, Sanad, Ammar, & Ibrahim used the genetic algorithms taking into account safety conditions and environmental considerations, as well as improving the model to measure the actual travel distance [28]. Also, Lam, Ning and Lam introduced a modified genetic algorithm using the Min-Max Ant System (MMAS) to develop a better initial population for the problem, instead of using randomly generated randoms in the traditional genetic algorithm theory [29]. Recently RazaviAlavi & AbouRizk used an integrated simulation model based on modified GA to optimize site layout planning by reducing the overall layout cost [30].

Data representation in optimization models is of two types, discrete or continuous [31]. Most of the models adopted the discrete formulation of the data to determine the locations of the temporary facilities due to its little time consumption and the ease of its calculations. While others adopted the continuous formulation. The discrete formulation takes only certain values for the positions of temporary facilities knowing that these values can be infinite, but each is distinct and no grey area in between. As an example of discrete representation models, Al Hawarneh, Bendak, & Ghanim planned the site space using a grid layout so that the sites available for allocation are grids' units, and there is no chance for a facility to occupy the position between two grids. the user controls the number of positions available for allocation by defining grid size [31]. On the contrary, in case of continuous formulation, the facility can occupy any position over a continuous range between specified discrete values. There is an infinite number of values between any two values over a continuous range. For example, Xu & Song

used a coordinate system to model the site space where any point in this system can be an option for allocation [32].

In this research, a 4D-BIM based dynamic site layout planning (4D-BIM DSLP) model is developed. In contrast to most previous studies that restricted their dynamic models to the progress in the construction schedule, our research aimed to present a 4D-BIM dynamic model for site layout planning that considers the progress in the construction phases as well as the delays and float time that construction may encounter. The model was also formulated in a continuous manner, where no fixed number of potential locations within the site area is assumed. We further utilized a novel framework based on Python to solve the optimization problem using genetic algorithms.

This research seeks to address two main questions: (1) How can float time be effectively integrated into site layout planning? (2) How can continuous layout optimization enhance facility placement efficiency using 4D-BIM and genetic algorithms?

3 Research aims & methodology

Previous studies have shown that dynamic models for construction site layout planning (CSLP) often fail to account for potential delays in some activities within the allowable time frame. As a result, such delays may necessitate changes in the locations of temporary facilities. Moreover, most existing models assume a discrete placement of facilities at predefined points.

This research aims to develop a dynamic construction site layout model that considers possible activity delays within the permitted time frame (float times), while also supporting the continuous allocation of temporary facilities within the site area, in order to improve simulation efficiency. To achieve this objective, as in Fig. 1 the following methodology was adopted:

1. Literature Review.

- Review of previous CSLP models.
- Analysis of related problems and limitations.

2. Problem Analysis.

- Identify and discuss the main features of CSLP.
- Highlight key challenges and influencing factors.

3. Model Development.

- Integrate the BIM-based planning model with the project schedule.

CSLP Model Development and Validation Workflow

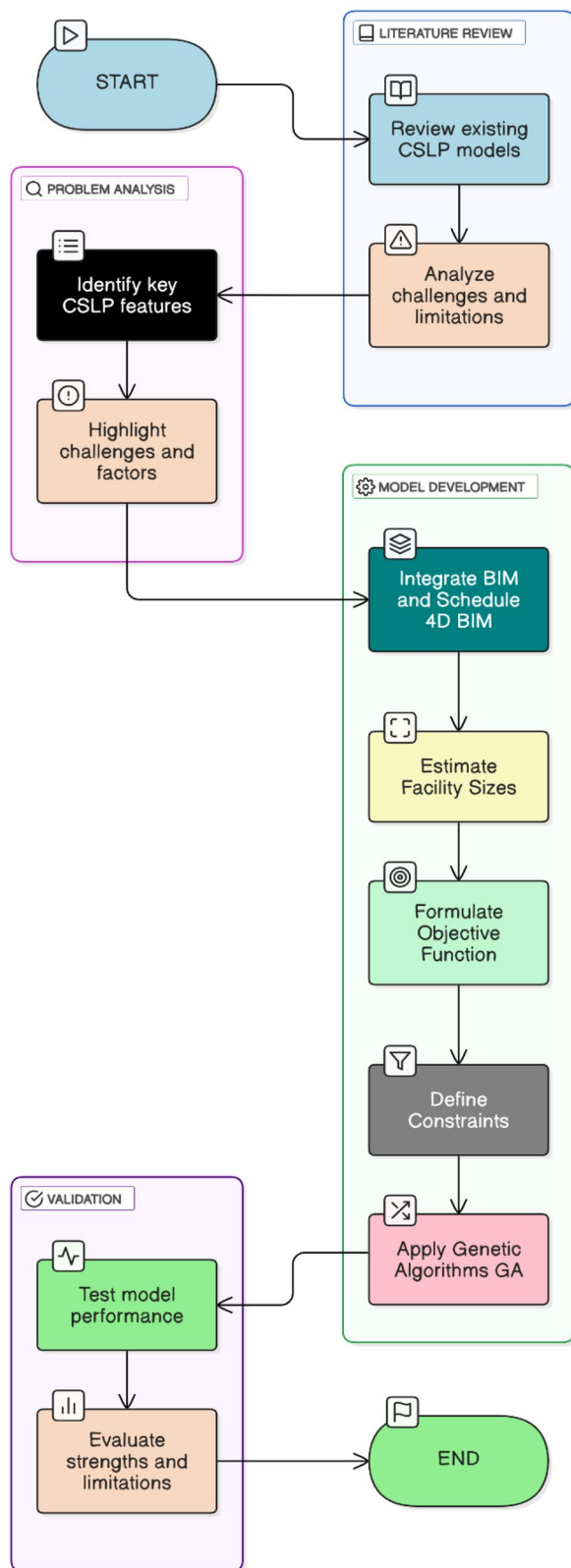


Fig. 1 Research methodology flowchart

- Estimating facility size using 4D-BIM technology.
- Defining the objective function.
- Defining limitations in site layout planning for construction projects.
- Solving optimization problems with Genetic Algorithms.

4. Validation.

- Test the validity of the proposed model.
- Evaluate its strengths and identify limitations.

4 Dynamic layout planning considering float times

A construction project begins in different phases or stages of construction. Piling, framing, mechanical, electrical and plumbing (MEP) work and finishing are common examples of different stages. As a result, the project needs of temporary facilities change according to each stage, for example, the facilities required in one stage may not be required in the next stage. As a result, unnecessary facilities are dismantled and their places are occupied by new ones. Sometimes the use of some facilities is changed to avoid the costs of disassembly and installation. To model this ever-changing facility requirement, dynamic planning models must be created. Dynamic layouts are set of partial layouts distributed over certain time periods, so that the total of them represents the entire construction site layout [5]. In most of the methods, the construction project is divided into phases, and for each phase partial layouts are defined [33]. These partial layouts must be interconnected as some facilities are required in multiple phases (see Fig. 2).

In this paper, we develop an automatic method using the construction schedule to create a dynamic model for site layout planning, where the activities of the construction schedule are linked with the facilities required for each activity. For example, the first floor slab activity was linked with both rebar stores and blacksmithing workshops, which represent the facilities required to store and process the materials used in the first floor slab activity. Where the facilities required for each activity are first defined by forming the so-called facilities requirements schedule. The rationale for linking activities to temporary facilities is that these facilities will be present on the site as long as the activities that require them are in progress, and therefore each bar in the facilities requirements schedule represents the total duration that the facility will occupy the construction site. In general, the materials required for each activity will be delivered two days before the date of their actual need at the site in order to avoid delays in delivery, as this point must be taken into account when creating the facilities requirements schedule.

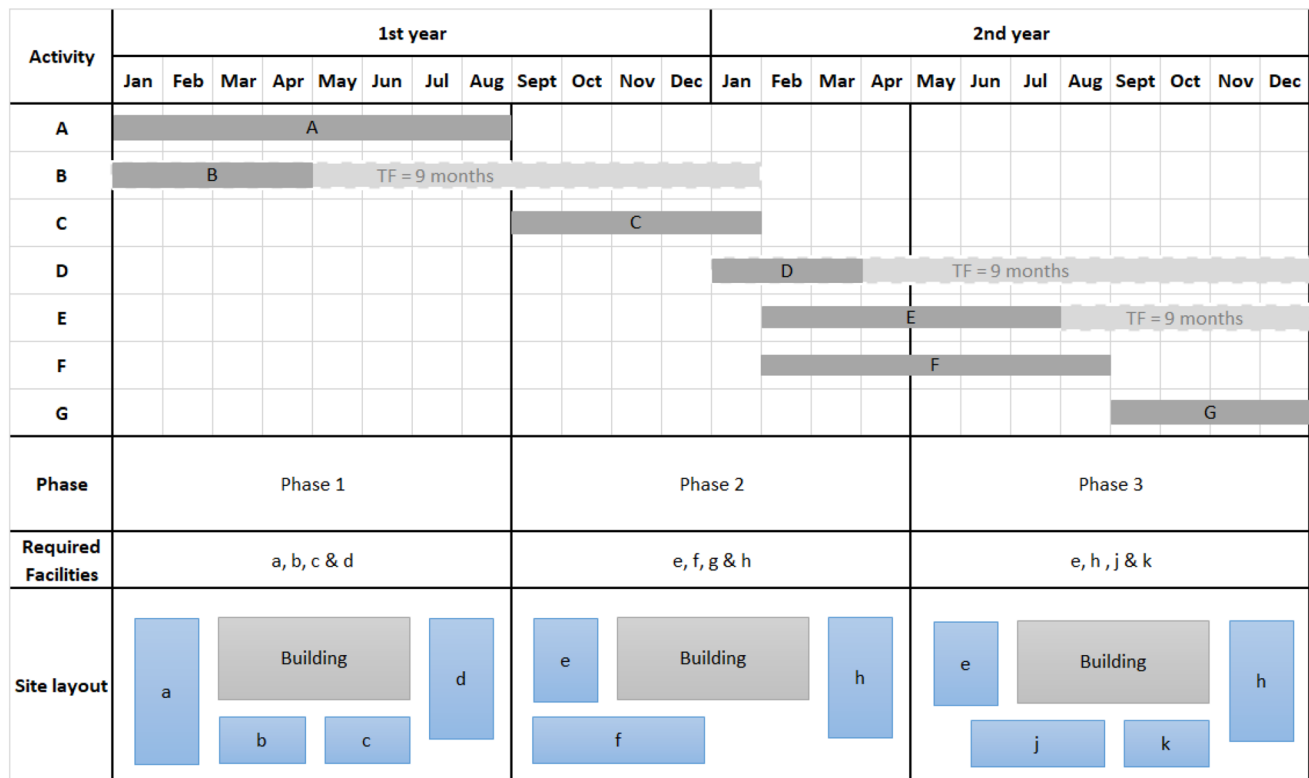


Fig. 2 Dynamic CSLP based on construction schedule

Since most scheduling softwares allow the schedule to be exported as a csv file, we were able to use Microsoft Excel to determine the facility requirements table from the activity schedule (see Fig. 2).

A construction project consists of a large number of activities arranged logically according to the construction sequence. Float times are usually scheduled between activities to set the pace of construction without delaying the project beyond its deadline. Float time assigned to one activity affects float times for all activities on the same network path. In other words, if an activity is delayed beyond its early end date it affects the float times of all activities on the same network path. Thus, a group of activities scheduling appears in which the start and end dates of the activities differ without affecting the deadline of the project. This study traced the effect of float time on construction site layouts for all possible construction plans. According to Fig. 3, in the example project seven activities are scheduled (activities A through G). The Gantt chart shows the beginning and end of each activity. The critical path for an example project is $A \rightarrow C \rightarrow F \rightarrow G$. The activities of this path must remain on schedule so that the project duration is not affected. Activities B, D, and E all have float times, where these activities can be moved within the limits of float times without affecting the overall project time. The effects of float times on these three

activities and their effect on the dynamic layout of the site were analyzed.

5 4D-BIM based facility size estimation

The solution of CSLP problem depends on optimizing the locations of temporary facilities within the construction site. Temporary facilities are treated as rectangles with dimensions that vary according to the nature of the materials stored within them. Most CSLP models require each facility to be dimensioned by the user before any optimization can be performed, but this approach has a major drawback, as it leads to poor optimization efficiency. Temporary facilities are used to store both equipment and materials necessary for construction, as well as to provide workspace for personnel within the construction site, so facilities should be large enough to meet these requirements. For example, the layout planner can specify an area of 100 m^2 for a site office that contains 10 people.

However, it is not possible to accurately determine the dimensions of the temporary facilities except after taking into account the space available within the construction site for the establishment of these facilities. In this case, facilities can take a variety of length and width values to eventually meet the necessary floor space for each facility, taking

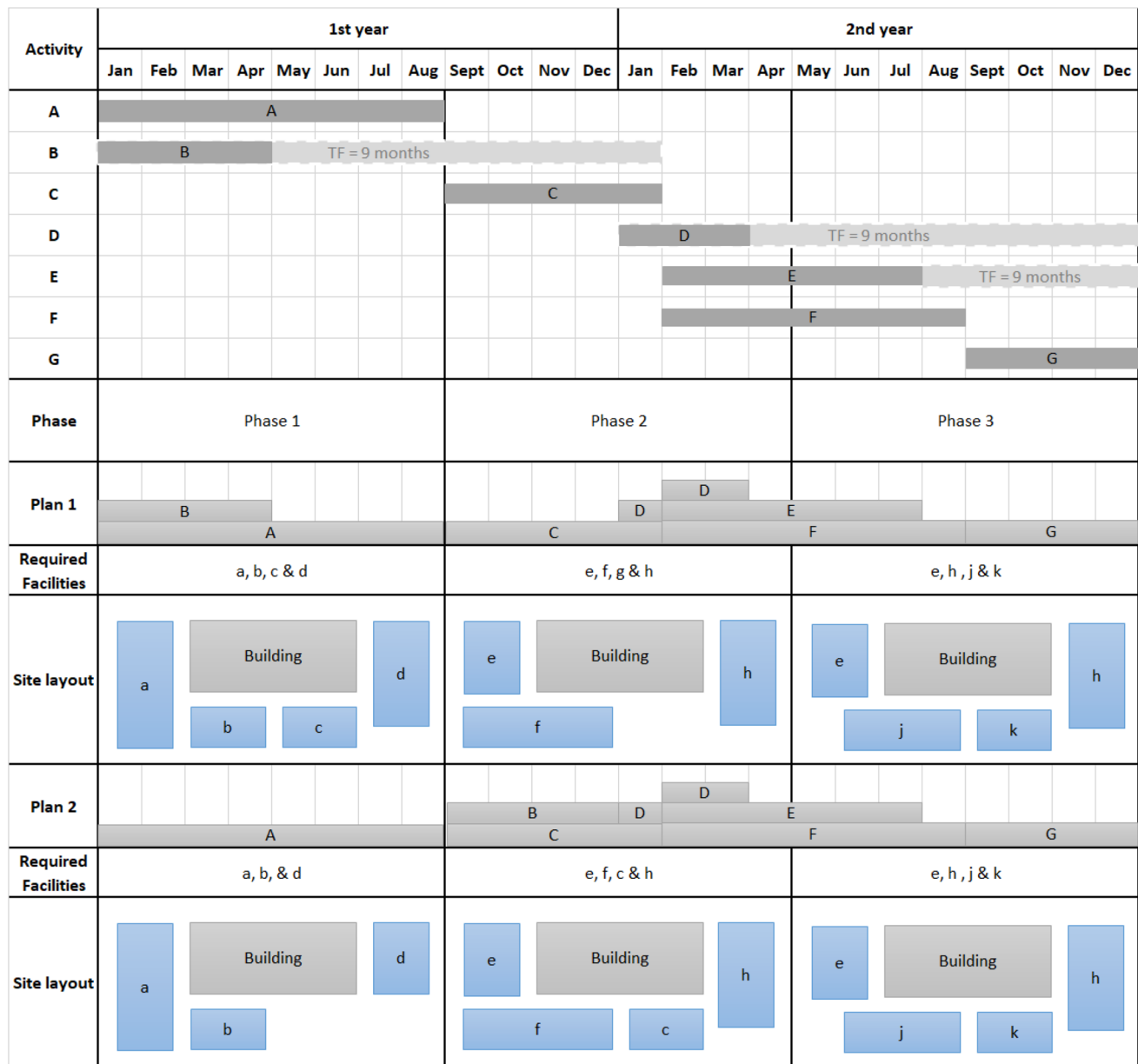


Fig. 3 Dynamic CSLP considering float times

into account that the length and width values are appropriate to the nature of the materials stored within each of them. Hence, defining fixed dimensions for each facility before locating each facility severely limits the range of possible solutions. Therefore, it is necessary to optimize the dimensions of each facility simultaneously with optimizing the locations of each of them, especially in construction sites with limited space. In this research we consider facility dimensions to be optimized by genetic algorithms.

Another drawback of defining the dimensions of the temporary facilities in advance by the user is that it makes the process very slow and tedious, especially if a change occurs in the designs, as it requires recalculating the dimensions

of the temporary facilities by the user every time a change order occurs instead of doing this task automatically, which leads to more work and effort for the layout planner. In this paper, we propose a methodology that uses information from the BIM model to determine facility sizes as shown in Fig. 4. In the first step, we use the methodology of estimating the required area of each facility based on BIM model. These areas are then used as a basis for determining their optimal dimensions.

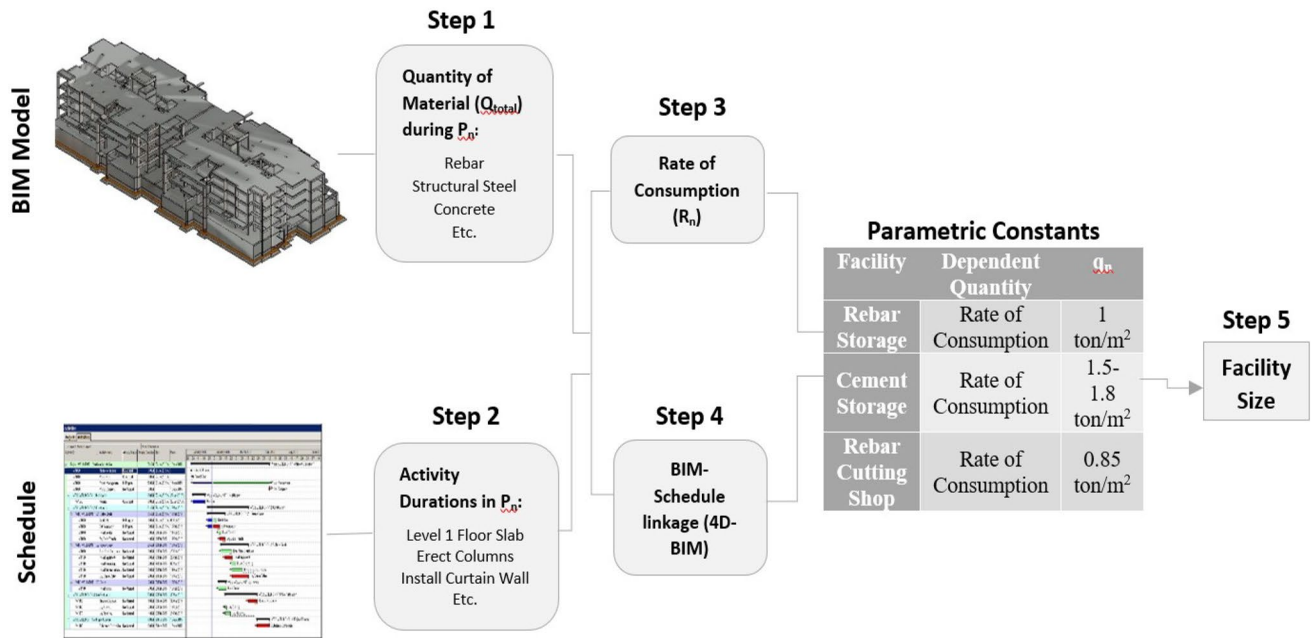


Fig. 4 The proposed facility size estimation module

5.1 Facility area estimation

The main function of temporary facilities is to provide support for construction processes. Temporary facilities can be classified into 3 categories: (1) Storage facilities, to temporarily store materials necessary for construction such as steel storage yards, cement warehouses, as well as equipment storage yards, where storage facilities must provide complete protection for materials from damage and exposure to the sun and weather factors, with a sufficient area for the materials inside. (2) Processing facilities, for processing raw materials within the construction site, such as forging workshops, steel bending yards, and rebar cutting. Processing facilities must provide sufficient space for personnel and equipment to work together safely and effectively. (3) Accommodation facilities, which provide shelter for individuals and meet the requirements of the participants in the construction project, such as site offices, storage rooms and restrooms. These facilities must provide enough space for people to work efficiently. Thus, the size of the storage facilities is related to the quantities of materials to be stored, while the size of the processing and accommodation facilities depends on the number of machines and personnel, respectively.

Considering constant quantity storage cycles, the maximum quantity allowed to be stored is related to the consumption rate of that material, and the time between two successive deliveries as follows:

$$Q_{mn} = R_{mn} \times T \times K \quad (1)$$

where m stands for m^{th} plan, n stands for the n^{th} period (P_n), Q_{mn} is the maximum stock level in the n^{th} period of the m^{th} plan, R_{mn} indicates the peak demand for materials or the rate of consumption of that resource in the n^{th} period of the m^{th} plan, and T is the time between successive deliveries, K denotes the consumption fluctuation factor (1-1.3) [34]. Thus, the size of the storage facility can be calculated by knowing the rate of consumption of the material inside it, and the time between two successive deliveries as follows:

$$As_{mn} = \frac{Q_{mn}}{I S q_n} \quad (2)$$

where As_{mn} denotes the area of the storage facility in the n^{th} period of the m^{th} plan, q_n is a constant and is an indicator of the amount of material that can be stored in the unit space (quantity/m²), S denotes the number of stacking layers, I denotes the material utilization index (to consider losses Materials, $I \leq 1$).

In the case of processing facilities, the number of machines used depends on the amount of material to be processed per day. For example, the number of machines necessary for the rebar bending workshop is determined by the amount of rebars to be processed every day, as well as the capacity of the machine. Hence the size of the processing facility can be determined as follows:

$$Ap_{mn} = K \times R_{mn} \quad (3)$$

where Ap_{mn} denotes the area of a processing facility in the n^{th} period of the m^{th} plan, K is a constant depends on the

material to be processed, and R_{mn} denotes the consumption rate of the dependent resource in the n th period of the m th plan.

Since no information about number of personnel through BIM model, the size of accommodation facilities must be defined by the user manually. Based on this logic, the size of the facility can be calculated in four steps as shown in Fig. 5. One of the drawbacks of the proposed framework is assumption that materials will be delivered to the construction site at fixed periods (T).

The first step: Extraction of materials quantities framework. Where the quantities of materials used in each structural element are calculated through the BIM model. For example, for “Column 001”, both the amount of concrete and the amount of reinforcing steel must be specified. This was performed by extracting the amounts of material from the individual components from the BIM model. Autodesk Revit was used as a modeling program that we rely on to extract and tabulate the materials quantities of different elements. We can also calculate other derived quantities such as scaffolds and formworks which are missed in the BIM model.

The second step: Calculating of the activity duration framework. In this framework, the duration of each activity is determined depending on the construction schedule. For example, the approximate time for the “Level 2: Columns 001–010” activity is determined from the project schedule.

The third step: linking the schedule to the BIM model. In this framework, the activities of the schedule are linked to the corresponding elements in the BIM model. We use Navisworks to link each element in the BIM model with the corresponding activity in the construction schedule. For example, the activity of pouring “Level 2: Column 010” concrete is linked to the corresponding element in the BIM

model to determine the amount of concrete used in intended column [35].

The fourth step: Calculate the Consumption Rate. We get the rate of material consumption during the various stages of construction by dividing the amount of material consumed by the consumption period. This approach forces a linear consumption rate for simplicity.

Step 5: Estimating the Facility Size. The size of each facility is calculated using the previously defined relationships after defining the time interval between two consecutive deliveries by the user.

5.2 Permissible facility dimensions

Once we have determined the floor area of the temporary facility, we have to determine its dimensions so that those dimensions take into account the facility’s capacity for objects stored within. For example, a rebar warehouse that contains bars of standard length of 8 m should have one of its dimensions at least 8 m. Thus, when determining the dimensions of the facility, the nature of the objects stored inside should be taken into account, in addition to meeting the floor area requirements of the facility. Based on this, a set of mathematical relationships were developed that represent the possible dimensions of the facility.

Suppose a rectangular facility has floor area A , length L , and width W . Suppose further that L_{min} and W_{min} are the minimum allowable length and width respectively, whose values are determined by the size of the objects stored inside the facility.

$$L = \alpha + L_{min}; \text{ Where } \alpha \geq 0 \quad (4)$$

$$W = \beta + W_{min}; \text{ Where } \beta \geq 0 \quad (5)$$

$$(L_{min} + \alpha) (W_{min} + \beta) = A \quad (6)$$

$$\beta = \frac{A - L_{min} W_{min} - \alpha W_{min}}{L_{min} \alpha} \quad (7)$$

For each value of α there is a corresponding value of β . This gives us a range of permissible values for both the length and width of each facility. Based on the characteristics of the site and the available spaces, the optimal dimensions of each facility are determined within the range of permissible values.

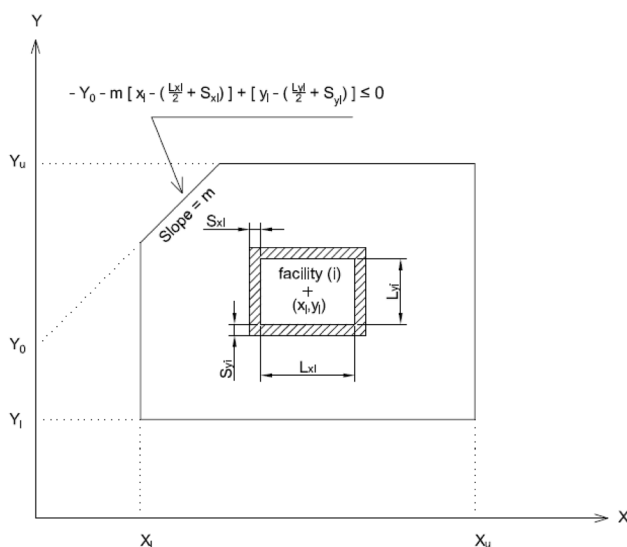


Fig. 5 Boundary constraints

6 Objective function formulation

After determining the available site area for placing temporary facilities and the requirements of each facility, we formulate the CSLP problem as an optimization problem. People, equipment, and materials move from one facility to another to perform a specific task, where each trip bears a cost directly proportional to the length of the trip and the method of transportation. The trip frequency is stored in the trip frequency matrix, and the trip cost per unit distance is stored in the cost matrix. The cost of the trip per unit distance, which expresses the ease of transportation, depends on the nature of the material being transported and the method of transportation. The Total Transportation Cost (TTC) is calculated as follows:

$$TTC = \sum_{p=1}^N \sum_{i=1}^{n-1} \sum_{j=i+1}^n d_{ijp} E_{ijp} f_{ijp} \quad (8)$$

Where N denotes the construction phases number, n denotes number of temporary facilities, and d_{ijp} , E_{ijp} and f_{ijp} denotes the travel distance, cost per unit distance, and frequency of transportation between facilities i and j through phase p , respectively. Value of d_{ijp} depends on the location of each facility and obstacles on the construction site. E_{ijp} and f_{ijp} depend on the mode of transportation and the number of trips to be taken, respectively. No longer needed Facilities on construction site are dismantled, relocated, or reused to store other materials. As a result, each facility has a cost associated with assembly, dismantle, and relocation, which will rely on the required labor and equipment. Therefore:

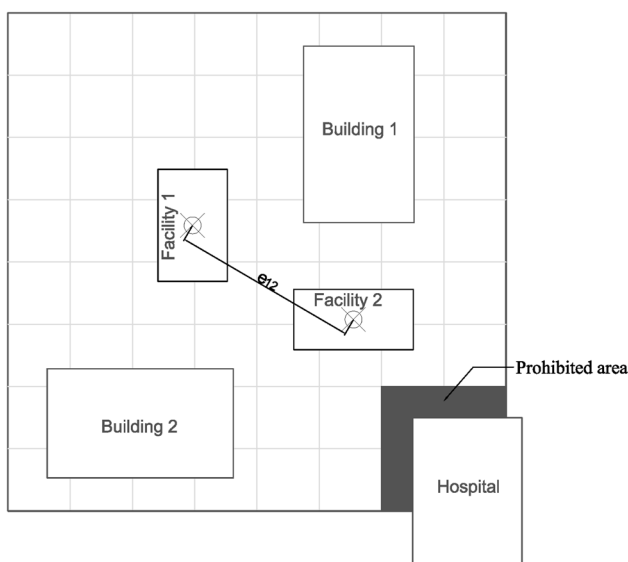


Fig. 6 Prohibited areas and minimum safe distance

$$FC = \sum_{p=1}^N \sum_{i=1}^n (AC_{ip} + DC_{ip} + RC_{ip}) \quad (9)$$

Where FC refers to the total cost due to assembling, dismantling and relocating the facilities. AC_{ip} , DC_{ip} , and RC_{ip} are the costs of assembling, dismantling, and relocation of facility i in phase p , respectively. The total cost of transportation, dismantling, moving and assembly can be represented mathematically as follows:

$$TC = TTC + FC \quad (10)$$

In other meaning, the optimization objective is to reduce TC , taking into account various constraints. Therefore, the optimal solution of site layout problem are the layouts corresponding to the minimum objective function value.

7 Constraints of the facility layout optimization problem

7.1 Boundary constraints

For the facility i , it is verified that this facility is located within the boundaries of the site (defined through the initial site drawings) - (see Fig. 5) if:

In X-direction:

$$x_i + \frac{L_{xi}}{2} + s_{xi} \leq X_u \quad (11)$$

&

$$x_i - \left(\frac{L_{xi}}{2} + s_{xi} \right) \geq X_l \quad (12)$$

In Y-direction:

$$y_i + \frac{L_{yi}}{2} + s_{yi} \leq Y_u \quad (13)$$

&

$$y_i - \left(\frac{L_{yi}}{2} + s_{yi} \right) \geq Y_l \quad (14)$$

For oblique boundary:

$$-Y_0 - m \left[x_i - \left(\frac{L_{xi}}{2} + S_{xi} \right) \right] + \left[y_i - \left(\frac{L_{yi}}{2} + S_{yi} \right) \right] \leq 0 \quad (15)$$

Where (x_i, y_i) refers to the geometric center of the facility i . While l_x, l_y refers to the facility dimensions in X & Y directions respectively, X_u, X_l, Y_u, Y_l refers to the upper and lower boundaries in X & Y directions respectively. m refers to slope of the oblique boundary while Y_0 refers to y-intercept. s_x, s_y refers to the minimum distances required between the facilities i and j , or facility i and the site boundary in the X & Y directions respectively.

7.2 Overlap constraints

We use mathematical inequalities as a solution to the problem of overlapping facilities with each other or with the buildings under construction, where the facility is represented mathematically by knowing (1) the coordinates of the geometric center of the facility, and (2) the dimensions of the facility, and accordingly the constraint to avoid overlapping between facilities is:

In X-direction:

$$|x_i - x_j| \geq \left(\frac{l_{xi}}{2} + \frac{l_{xj}}{2} \right) + \max \{s_{xi}, s_{xj}\} \quad (16)$$

In Y-direction:

$$|y_i - y_j| \geq \left(\frac{l_{yi}}{2} + \frac{l_{yj}}{2} \right) + \max \{s_{yi}, s_{yj}\} \quad (17)$$

7.3 Safety constraint

7.3.1 Minimum safe distance

Sometimes we have to leave a minimum separating distance between a particular hazardous facility and other nearby facilities to avoid bad effects on users of other facilities. In our research, we represent the safety constraint as following:

$$e_{ij} \geq s_{ij} \quad (18)$$

$$e_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (19)$$

Where e_{ij} denotes the Euclidean distance between facilities i and j , s_{ij} is the minimum safe distance between facilities i and j , (x_i, y_i) and (x_j, y_j) are centroids of facilities i and j respectively. Large values of s_{ij} indicate safety concerns when the facilities i and j are placed in close to each other.

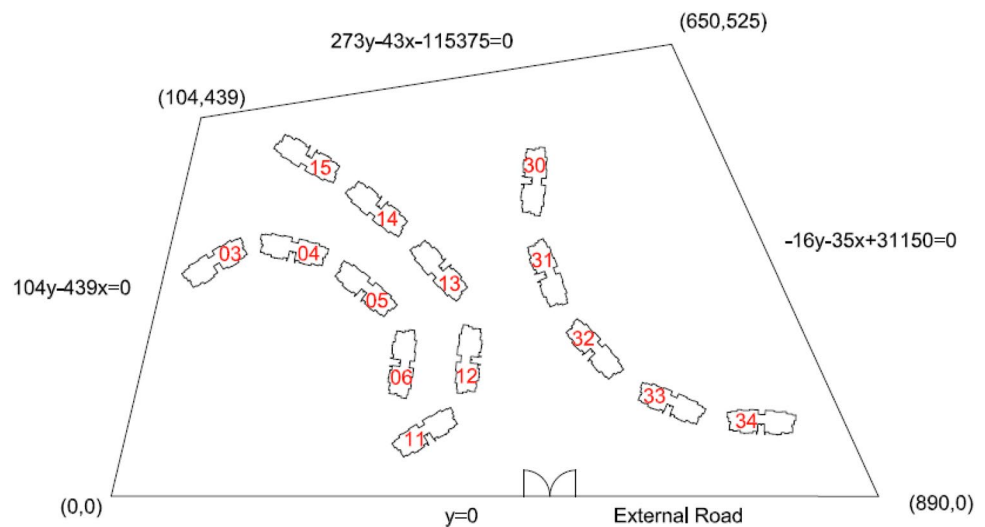
7.3.2 Prohibited areas

It is important to allocate pre-defined areas in the work site so that it is forbidden to place any temporary facilities in them. These areas are near sensitive entities such as schools and hospitals to prevent facilities with a harmful impact such as noise and air pollutants from affecting these sensitive entities (See Fig. 6).

7.3.3 Safety zones

Safety zones are additional zones that are added to the area of the facility to avoid the negative effects and injuries that may arise from falling objects or flying materials as a result of various operations. Safety zones must be appropriately defined in relation to relevant regulations, such as UBC

Fig. 7 General plan of the project example



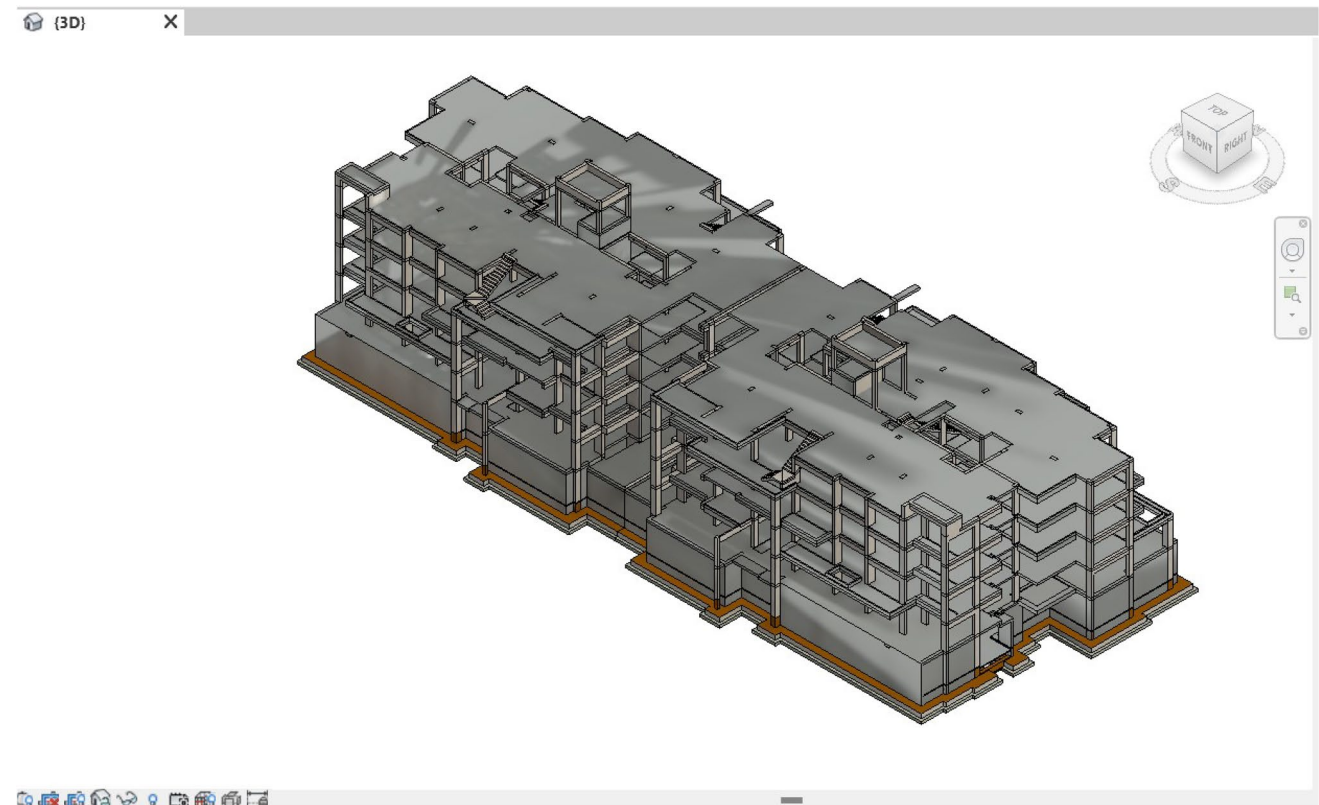


Fig. 8 Revit model of the typical residential building in the project example



Fig. 9 Primavera P6 Schedule of project example

Table 1 Facilities needed for the site layout

Facility number	Name	Type	Area (m ²)	Phase required	Size calculation
1	Tool shed	Storage	55	1,2,3	By user
2	Site offices	Residence	70	1,2,3	By user
3	First aid	Residence	50	1,2,3	By user
4	Restrooms	Residence	35	1,2,3	By user
5	Labors rest area	Residence	100	1,2,3	By user
6	Parking	Residence	150	1,2,3	By user
7	Scaffold storage yard	Storage	40	1,2,3	BIM & Schedule
8	Rebar storage	Storage	70	1,2,3	BIM & Schedule
9	Formwork storage	Processing	50	1,2,3	BIM & Schedule
10	Aggregate warehouse	Storage	60	1,2,3	BIM & Schedule
11	Cement warehouse	Storage	35	1,2,3	BIM & Schedule
13	Plumbing storage	Storage	50	1,2,3	BIM & Schedule
14	Electrical systems storage	Storage	75	1,2,3	BIM & Schedule
15	Batch plant	Processing	100	1,2,3	BIM & Schedule
16	Welding shop	Processing	75	1,2,3	BIM & Schedule

(1985) and OSHA (1987). For example, if the actual area of the facility = 50 m² and the appropriate safety area = 20 m², then the area of the facility = 50 + 20 = 70 m².

7.3.4 Another regulations and safety standards

Another safety rules have also been compiled from the Safety and Health Administration (OSHA 1987), Safety and Health Manual (ILO 1992), Building Codes (UBC 1985; SBC 1985). Some of these regulations are listed below.

- A clearance of at least 10 feet from buildings or structures must be maintained for piles of wood and other combustible materials for use in construction.
- Aisles around and between combustible storage piles in an open courtyard should be 15 feet as minimum and free of litter accumulation.
- Taking into account that the materials stored inside buildings under construction are not within 10 feet of an external wall whose height does not exceed the height of the stored materials, and that these materials are not stored within 6 feet of a crane or internal floor openings.

7.4 Tower crane reachability constraint

Some material stores should be located at a distance not exceeding the radius of the tower crane. This can be mathematically represented as follows:

$$(x_r - x_f)^2 + (y_r - y_f)^2 \leq T_R^2 \quad (20)$$

Where (x_r, y_r) represents the tower crane's axis of rotation, (x_f, y_f) refers to the farthest corner of the facility from the tower crane, and T_R refers to the radius of the tower crane. The pickup radius of crane is inversely related to the weight of the object to be lifted, so heavy objects should be placed closer to the crane, while lighter objects can be stored farther away. This model is based on the user defining the crane

Table 2 Facilities and minimum dimensions required for each

Facility number	Name	Type	Area (m ²)	L_{min} (m ²)	W_{min} (m ²)	α
1	Tool shed	Storage	55	4	3	1
2	Site offices	Residence	70	3	2.5	1
3	First aid	Residence	50	3	2.5	1
4	Restrooms	Residence	35	3	2.5	1
5	Labors rest area	Residence	100	3	2.8	1
6	Parking	Residence	150	3	2	1
7	Scaffold storage yard	Storage	40	2.5	2.5	1
8	Rebar storage	Storage	70	13	3	1
9	Formwork storage	Processing	50	4	2	1
10	Aggregate warehouse	Storage	60	2	2	1
11	Cement warehouse	Storage	35	2	2	1
12	Plumbing storage	Storage	50	4	1.5	1
13	Electrical systems storage	Storage	75	2	1.5	1
14	Batch plant	Processing	100	3	3	1
15	Welding shop	Processing	75	4	2	1

```

1 import random
2 # import sys
3 import math
4 import csv
5
6 s = int(input("please enter the number of solutions (size of Population): ")) # 1000
7 n = int(input("please enter the number of site facilities: "))
8 child_num_generated_from_cross_over_two_parents = int(
9     input("please enter the number of child generated from crossover process: "))
10 num_generations = int(input("please enter the number of generations: "))
11
12 all_data_population = []
13 transportation_cost_population = []
14 best_parents_and_fitness = []
15
16 center_area_under_construction_x = float(input("please enter the center of total area under construction x: "))
17 center_area_under_construction_y = float(input("please enter the center of total area under construction y: "))
18

```

C:\Users\LENOVO\PycharmProjects\omar\venv\Scripts\python.exe C:/Users/LENOVO/PycharmProjects/omar/main.py
 please enter the number of solutions (size of Population): 1000
 please enter the number of site facilities: 10
 please enter the number of child generated from crossover process: 1000
 please enter the number of generations: 200

Fig. 10 Python code to solve site layout problems using genetic algorithms

location, and the allowable lifting radius for each stored material.

8 Optimization using genetic algorithms

CSLP problem is considered to be difficult and many studies have sought to arrive to solutions using heuristics [36–40] or mathematical optimization [41] techniques. Genetic Algorithms (GA) is one of the most common methods for solving site layout problems due to its ease of implementation. Genetic algorithms combine the genes of two elements (two solutions) from the same generation that we call (parents) to produce a third solution that is evaluated later based on the objective function [42]. The third solution is kept if it is better than its parents, otherwise it is discarded. This process is repeated on subsequent generations to produce new generations that are more fit than the previous generations until the solutions converge on the optimal solution and the production of new generations does not represent a significant difference in terms of fitness. Mutation operations are developed to prevent the algorithm from getting stuck at local minimums.

8.1 Gene encoding

In Genetic Algorithms, the variables to be optimized are regarded as genes. Traditional CSLP approaches only improve facility locations on the construction site. Accordingly, facilities locations are encoded into genes. In this

paper, we look at optimizing the location of the facilities in conjunction with optimizing their dimensions, so the dimensions of the facilities are also encoded.

An array of the real values of coordinates and facilities dimensions is used as a gene in this work. Suppose there are a number p of parent genes:

$$P_1 = (F_{11}, F_{12}, F_{13}, \dots, F_{1n})$$

Where:

$$F_{11} = (x_{11}, y_{11}, l_{11}, w_{11}), F_{12} = (x_{12}, y_{12}, l_{12}, w_{12}), \\ F_{13} = (x_{13}, y_{13}, l_{13}, w_{13}), F_{1n} = (x_{1n}, y_{1n}, l_{1n}, w_{1n})$$

And:

$$P_2 = (F_{21}, F_{22}, F_{23}, \dots, F_{2n})$$

Where:

$$F_{21} = (x_{21}, y_{21}, l_{21}, w_{21}), F_{22} = (x_{22}, y_{22}, l_{22}, w_{22}), \\ F_{23} = (x_{03}, y_{23}, l_{23}, w_{23}), F_{2n} = (x_{2n}, y_{2n}, l_{2n}, w_{2n})$$

$$P_p = (F_{p1}, F_{p2}, F_{p3}, \dots, F_{pn})$$

Where:

$$F_{p1} = (x_{p1}, y_{p1}, l_{p1}, w_{p1}), F_{p2} = (x_{p2}, y_{p2}, l_{p2}, w_{p2}), \\ F_{p3} = (x_{p3}, y_{p3}, l_{p3}, w_{p3}), F_{pn} = (x_{pn}, y_{pn}, l_{pn}, w_{pn})$$

Table 3. Part of results obtained from python code in the form of csv file

Generation	Facility 1				Facility 2				Facility 3			
	x	y	l	w	x	y	l	w	x	y	l	w
1	514.47	317.91	14.00	3.33	715.34	167.97	10.00	4.64	145.90	3.95	5.00	8.75
2	577.91	234.24	14.00	3.33	702.77	172.77	11.54	4.07	139.88	255.04	5.00	8.75
3	577.80	236.16	14.00	3.33	708.08	177.74	11.30	4.34	123.76	200.40	5.00	8.75
4	577.82	236.80	14.00	3.33	707.16	175.53	11.24	4.34	123.24	191.89	5.00	8.75
5	577.81	236.57	14.00	3.33	707.34	176.23	11.24	4.33	123.20	192.15	5.00	8.75
6	577.81	236.59	14.00	3.33	707.48	176.38	11.24	4.33	123.18	192.13	5.00	8.75
7	577.82	236.59	14.00	3.33	707.43	176.54	11.24	4.34	123.18	192.13	5.00	8.75
8	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
9	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
10	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
11	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
12	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
13	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
14	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
15	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
16	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
17	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
18	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
19	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75
20	577.82	236.59	14.00	3.33	707.43	176.59	11.24	4.34	123.19	192.13	5.00	8.75

where: F_{pn} is an array of x , y coordinates as well as dimensions values (l , w) for all the facilities of gene p , while n refers to numbers of facilities to be laid out.

8.2 Modified crossover and mutation operators

8.2.1 Crossover operator

Crossover operator is to make a child gene C from parents P_1, P_2 , where:

$$C_c = (F_{c1}, F_{c2}, F_{c3}, \dots, F_{cn})$$

Where:

$$F_{c1} = (x_{c1}, y_{c1}, l_{c1}, w_{c1}), F_{c2} = (x_{c2}, y_{c2}, l_{c2}, w_{c2}), \\ F_{c3} = (x_{c3}, y_{c3}, l_{c3}, w_{c3}), F_{cn} = (x_{cn}, y_{cn}, l_{cn}, w_{cn})$$

Where C_c is calculated as follow:

- Step 1: Randomly generate an integer number (m) which represent number of facilities to be crossed over, where: $1 < m \leq n$.
- Step 2: Randomly generate (m) integer numbers ($i_1, i_2, i_3, \dots, i_k$), where i_k represent the order of facilities to be crossed over ($1 < k \leq m$ & $1 < i_k \leq n$).
- Step 3: For $j = i_k$, generate $F_{cj} = (x_{cj}, y_{cj}, l_{cj}, w_{cj})$, where:

$$x_{cj} = \alpha x_{1j} + \beta x_{2j} \quad (21)$$

$$y_{cj} = \alpha y_{1j} + \beta y_{2j} \quad (22)$$

$$l_{cj} = \alpha l_{1j} + \beta l_{2j} \quad (23)$$

$$w_{cj} = \alpha w_{1j} + \beta w_{2j} \quad (24)$$

Where α and β , are real numbers generated randomly. The rest of the facilities have not been crossed over are just copied from P_1 .

If the position of P_1 and P_2 is exchanged, in a similar manner, application of the algorithm above can make another child gene.

In this crossover, the coordinate & dimensions values of child genes are obtained by arithmetic combination of parent genes coordinate & dimensions values. If $\alpha + \beta = 1$, then the crossover operator above is a weighted average of two parent genes and will make the corresponding coordinate & dimensions values of the child chromosome stay between its two parents. Specifically, if α & β is constants for both x and y coordinates and their sum equals one ($\alpha + \beta = 1$), then the children produced will be on the line that links between

the two parents. If there are restrictions on any coordinates, a typical process is applied to ensure that the coordinates and dimensions do not violate any of these restrictions.

8.2.2 Mutation operator

In this operator, a random number is added to the coordinates of a facility or its dimensions or both of them which randomly selected from the P chromosome, so that the original position of this facility or its dimensions changes. Another method can be applied such as randomly moving the locations of some facilities in the P chromosome instead of adding a random number to their coordinates. The same process can be applied to dimensions.

9 Demonstrative example

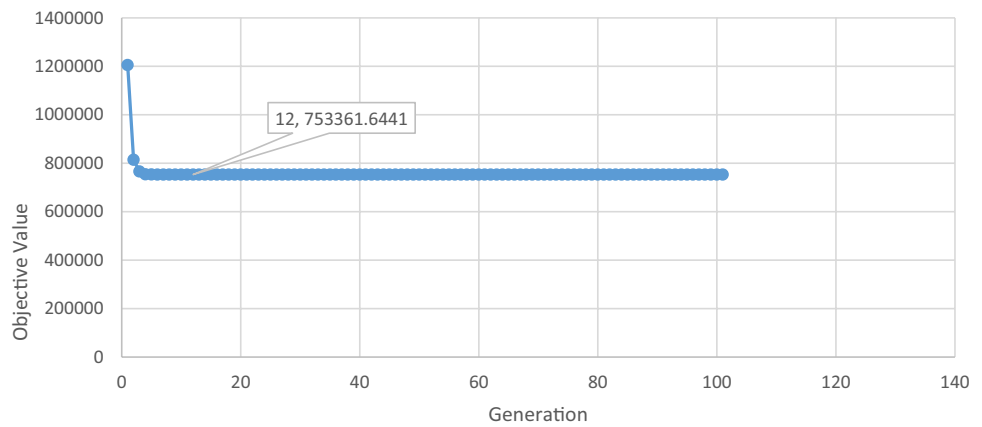
The developed model have been tested on an illustrative example of a real project. The project involves the construction of 14 typical residential buildings on 3 phases (See Fig. 7). Where the first phase consists of 5 Buildings (Buildings 30, 31, 32, 33 & 34), the second phase consists of 4 Buildings (Buildings 3, 4, 5 & 6), and the third phase consists of 5 Buildings (Buildings 11, 12, 13, 14 & 15), the site area is defined by the corner points (0, 0), (890, 0), (650, 525), (104, 439).

Autodesk Revit was used to create the BIM model (see Fig. 8). Primavera P6 was used to create a construction activities schedule (see Fig. 9). The schedule was exported from Primavera P6 as a csv file, which could be linked to the 3D-BIM model using Autodesk Navisworks to assign each schedule activity to the corresponding 3D-BIM Model element. A Revit API plugin was developed to determine the material quantities of different time periods, then we were able to determine the area required for each temporary facility, as well as the frequency of trips between facilities based on the amount of daily work. For simplicity we have assumed that each path starts and ends at the geometric center of the facility. The construction plan of the project requires sixteen temporary facilities for the first phase of the project. These facilities and related data are given in Table 1. L_{min} , W_{min} are assumed reasonably for each facility (See Table 2), For example, a rebar warehouse that contains bars of standard length of 12 m should have one of its dimensions at least 12 m. Thus, the dimensions of the facility are determined taking into account the nature of the things to be stored in it.

(Autodesk Revit 2018, <https://manage.autodesk.com/products/RVT?version=2024&platform=WIN64&language=EN>)

Table 4 Part of results obtained from python code in the form of Csv file

Generation	Facility_13			Facility_14			Facility_15			Fitness Value (total Cost)		
	x	y	l	w	x	y	l	w	x	y	l	w
1	393.52	154.42	4.00	18.75	559.46	270.19	7.00	9.58	711.49	186.98	7.00	7.08
2	503.19	146.04	4.00	18.75	548.57	243.97	8.55	8.63	670.35	152.39	7.83	5.23
3	541.37	146.29	4.00	18.75	545.47	224.91	8.36	7.58	646.81	153.26	7.89	5.19
4	535.61	146.84	4.00	18.75	547.15	216.86	8.37	7.29	649.23	153.18	8.03	5.12
5	537.24	146.75	4.00	18.75	547.13	216.03	8.37	7.32	649.56	153.10	8.02	5.14
6	537.26	146.78	4.00	18.75	547.12	215.99	8.37	7.38	649.52	153.12	8.02	5.14
7	537.25	146.78	4.00	18.75	547.12	215.86	8.37	7.38	649.53	153.12	8.02	5.14
8	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
9	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
10	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
11	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
12	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
13	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
14	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
15	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
16	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
17	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
18	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
19	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14
20	537.25	146.78	4.00	18.75	547.12	215.85	8.37	7.38	649.53	153.12	8.02	5.14

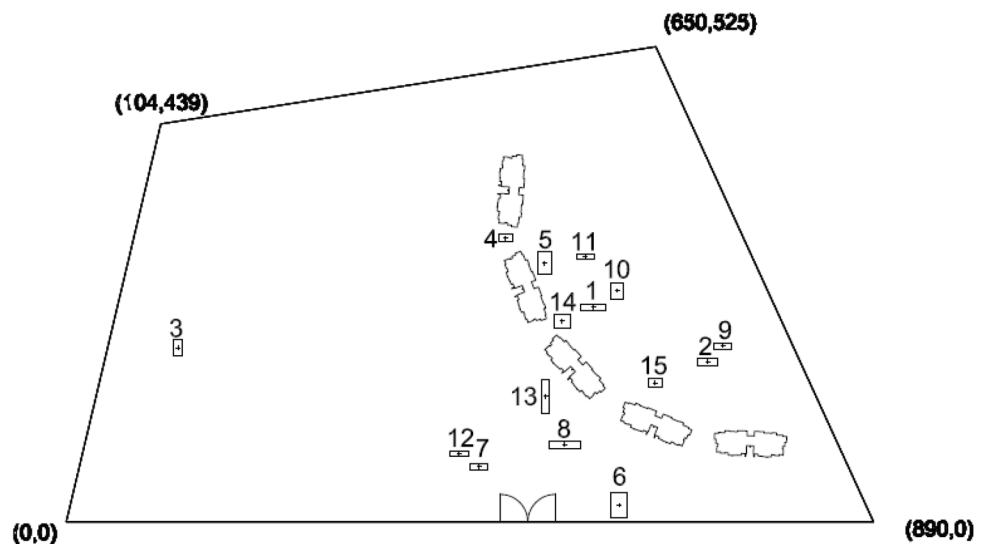
Fig. 11 Convergence of objective value**Table 5** Site layout of the optimal solution

Facility	x	y	l	W
1	577.8	236.6	14	3.3
2	707.4	176.6	11.2	4.3
3	123.2	192.1	5	8.8
4	484.7	313.4	7.4	3.7
5	527.3	285.7	7.7	12.9
6	538.6	417.9	9	14.5
7	455.3	61.1	9.9	3.5
8	549.5	85.0	17.1	4.4
9	724.0	194.2	9.3	3.5
10	598.8	249.2	7.2	8.9
11	572.6	293.1	9.5	2.8
12	433.3	75.1	10.8	2.6
13	537.2	146.8	4	18.8
14	547.1	215.9	8.4	7.4
15	649.5	153.1	8	5.1

9.1 Results summary

The results have been obtained in the form of csv file, (See Tables 3. and 4), Where the first row indicates the solution of the first generation due to crossover, while the solution of the second generation locates in the second row and vice versa. Each solution contains the facilities information, for example, facility 1 in the first generation has the geometric center (514.47, 317.91), and the length and width 14 m and 3.33 m, respectively. Convergence occurred at the 12th generation, where we obtained the lowest possible cost (753361 LE), while mating more does not give a significant cost reduction (See Fig. 11). The results of 12th generation have been extracted separately to represent the optimal solution (See Table 5), For example, facility (1) has the geometric

To solve this problem using genetic algorithms, a python code has been developed to solve such problems (See Fig. 10). Size of population has considered to be 1500, the children generated from crossover have considered to be 1000, while number of generations has been set to be 200.

Fig. 12 Geometric site layout of the optimal solution of 1st phase

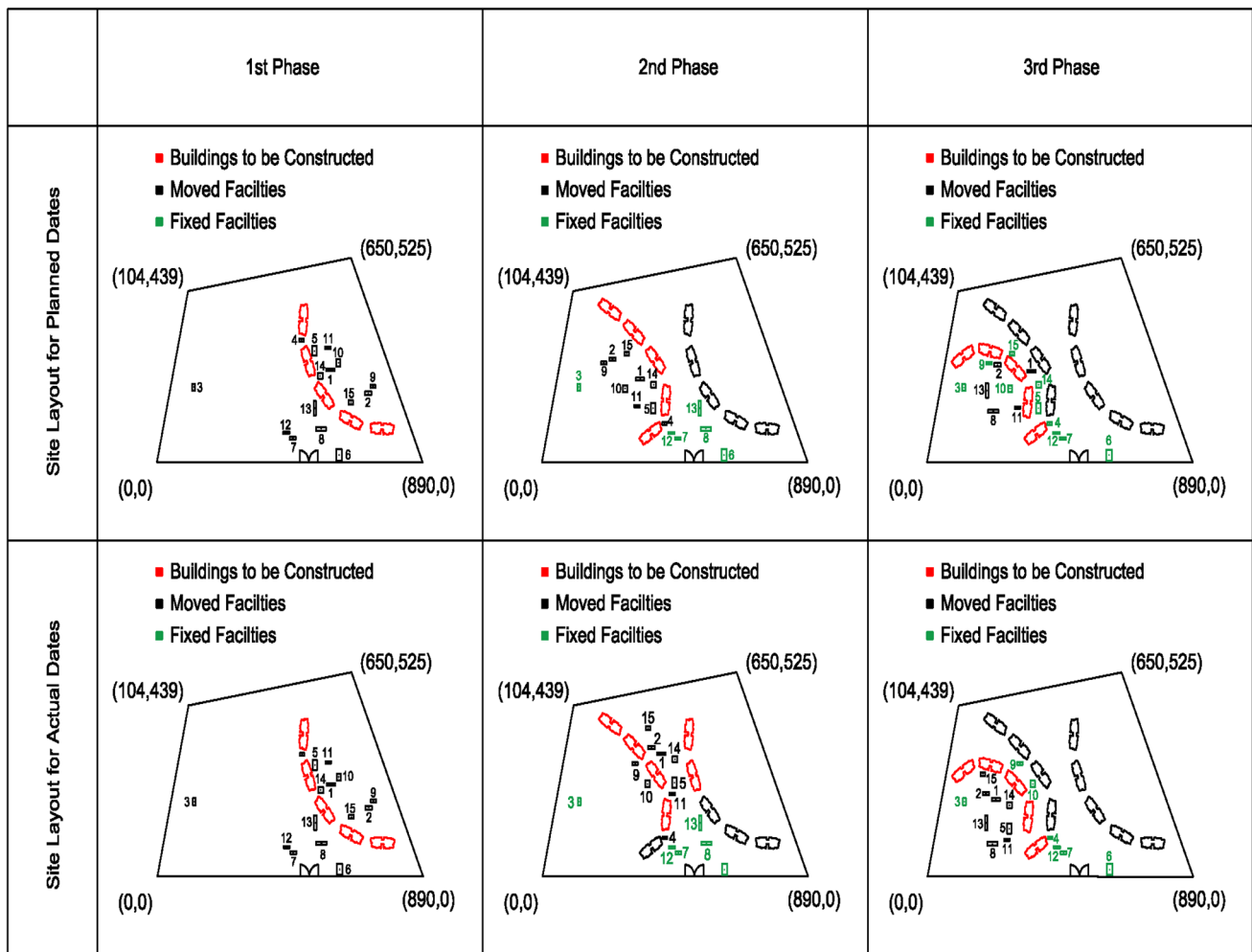


Fig. 13 Geometric site layout of the optimal solution of all phases (planned & actual dates)

center (577.8, 236.6) and the length & width 14 m, 3.3 m respectively. The temporary facilities have been laid on the site plan as shown in Fig. 12. Similarly for the other phases, facilities have been laid on the site plan for initial planned dates, then for final actual dates which reflects the dynamic changing nature of construction site layout (See Fig. 13).

10 Conclusion & future work

- In this research, a practical and efficient 4D BIM model for dynamic layout planning in construction sites is developed. The developed model can be used to optimize the placement of temporary facilities on construction site so as to maximize site efficiency and maintain site layout limitations. The dynamic nature of construction site layout due to transition from one construction phase to another or delaying and floating of schedule activities are considered both.
- Some concluding remarks are listed below:
 - The developed site layout planning system can help project managers laying out their sites in a dynamic way that consider both transition from one construction phase to another and delaying and floating of schedule activities.
 - The developed site layout planning system formulates site layout problem in continuous way where a coordinate system was used to model the space, and any point in this coordinate system might be an option for allocation.
 - BIM technology was used to facilitate the calculation of material quantities according to the change in the schedule and progress in construction and thus facilitate the dynamic planning of the site.
 - Genetic Algorithms is a powerful optimization method that can be satisfactorily used to solve site layout problems. GAs is generic and flexible, easy to be understood, need little information about the problem domain.

The developed site layout planning model can be extended to consider one or more of the following:

- Expanding the use of BIM technology to include visual modeling of construction sites and temporary facilities.
- Extend the objective function to account for secondary and intangible objectives.
- Enhance site layout planning by adding an option to optimize facilities material type (such as wood or bricks).
- Exploring the possibility of orienting the facilities at angles of 0, 90, 180 and 270 without significantly increasing the time required for the solution.
- Modeling accurately level difference between different site areas to account for increase and decrease of productivity of equipment.

Acknowledgements Not applicable.

Authors' contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis.

Funding This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability The datasets generated or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare that they have no competing interests.

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